

4.1

Master Data

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OBJECTIVE

This exercise teaches how to create a data flow, consisting of data sources, transformation, and data transfer process. This data flow is needed to load the master data from the data source into the Data Warehouse.

ACTIVITIES

Learn how to create a data flow object, data source, and transformation.

SOFTWARE PREREQUISITES

- Eclipse with SAP BW Modeling Tools
- SAP GUI

DATA SET

Global Bike (GB) data set

SYSTEM ACCESS

SAP BW/4HANA 2.0 from your UCC

Scenario

Jan Schmidt is Controlling Specialist at Global Bike, Germany. His boss, Hanan Cengiz, asks him to create the data model in the BW/4HANA 2.0 for Global Bike Germany and US with

the goal that Hanan can present their sales/cost analytics quicker to their CFO Wilhelm Ollenhauer. Jan has already modelled InfoObjects as elementary information objects in a preceding exercise. In this exercise Jan will load the master data into InfoObjects built previously. Migration of data into data warehouses is an important step. We usually accomplish this through ETL (extract, transform and load).

InfoObjects can be characteristics or key figures. Characteristics represent business entities like *product* or *customer* or attributes of business entities. The values of the business entities are kept in master data tables that “belong” to the InfoObjects.

Master data in BW consist of attributes (e.g. *Product Category* and *Division*) and a description (text). Since the description can be maintained in different languages it is stored in a separate table with an additional language key. Therefore, loading of master data is a two-step process: attributes and texts. Additional steps could be required to load more features of the InfoObjects, such as hierarchies and geographic data.

Master data are loaded separately from transaction data, as they are (more or less) stable over time. Transaction data typically contain a *combination of keys* of master data and several values of measures like revenue, etc. (e.g. we will load transactional data in a future exercise.)

It is possible to model the data flow in a graphical way. A data flow contains all required objects and their relationships for loading the data.

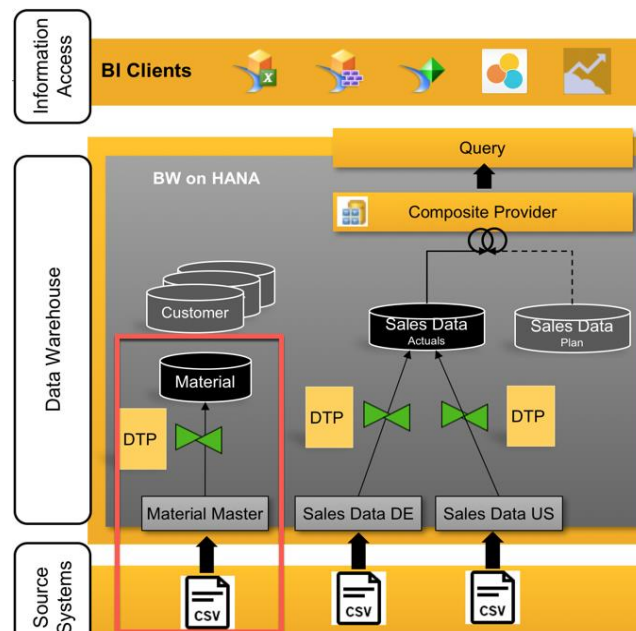


Figure 1

In this exercise you will upload *material* master data into the data warehouse. First, you will create a DataSource as a preparation for the data migration process. Then you will create a Data Flow between the *material* InfoObject (from the previous exercise) and the material master data DataSource. Before you can load the material attributes in the data warehouse,

you need to create a *transformation*, defines *rules*, and create *data transfer process*. After a load you will verify the material *attributes* have been successfully loaded. Then you will do the same thing for loading material *texts*.

The graphic below displays the complete process (10 tasks).

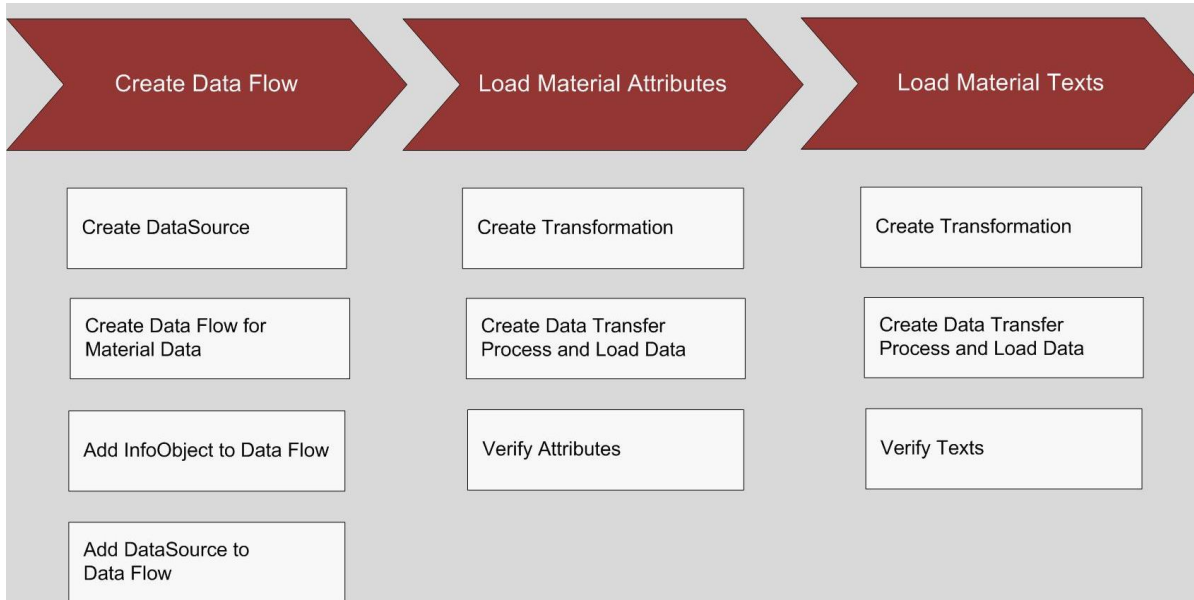


Figure 2

1. Create a DataSource for the Material Master Data

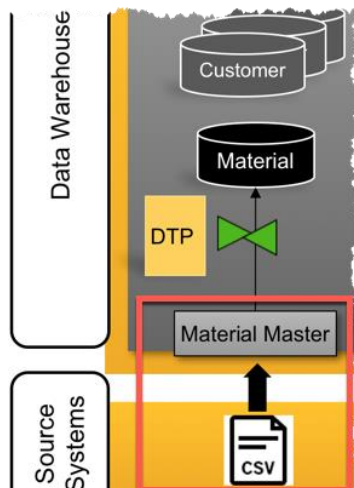


Figure 3

- a. Let's first explore the material master data file to get acquainted with the fields and formats of the data.

- b. Download the *Product.csv* (provided by your instructor) on your computer. Open it with a text editor to understand the structure of the data and the *separator* (;) being used. You will need this information when you create the DataSource.

```

1 MaterialDescr;MaterialNumber;ProductCategory;Division;Color;Components
2 Deluxe Touring Bike (black);DXTR1000;TOU;BI;Black;Shimano Road
3 Deluxe Touring Bike (silver);DXTR2000;TOU;BI;Silver ;Shimano Road
4 Deluxe Touring Bike (red);DXTR3000;TOU;BI;Red;Shimano Road
5 Professional Touring Bike (black);PRTR1000;TOU;BI;Black;Shimano XT
6 Professional Touring Bike (silver);PRTR2000;TOU;BI;Silver ;Shimano XT
7 Professional Touring Bike (red);PRTR3000;TOU;BI;Red;Shimano XT
8 Men's Off Road Bike Fully;ORMN1000;ORB;BI;;SRAM XO
9 Women's Off Road Bike Fully;ORWN1000;ORB;BI;;SRAM XO
10 Men's Off Road Bike Hard Tail (Shimano);ORHT1000;ORB;BI;;Shimano XT
11 Men's Off Road Bike Hard Tail (SRAM);ORHT2000;ORB;BI;;SRAM X9
12 Deluxe Road Bike (Shimano);DXRD1000;ROB;BI;;Shimano Ultegra
13 Deluxe Road Bike (SRAM);DXRD2000;ROB;BI;;SRAM Force
14 Professional Road Bike (Shimano);PRRD1000;ROB;BI;;Shimano Dura Ace

```

Figure 4

All ETL processes in BW need two objects:

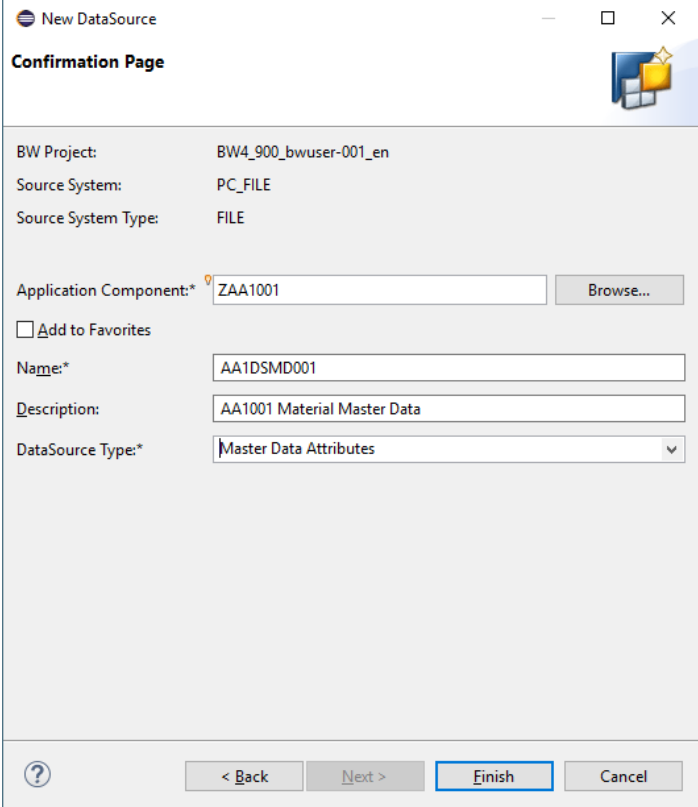
- A *Transformation* defines the rule set, how the fields are mapped between a source structure and a target structure such as an InfoObject
- The transfer of data between a source object and a target object is done by *Data Transfer Processes (DTP)*, which can be understood as a program that is scheduled and actually “does” the data transfer.

Sources, targets, transformations can be arranged in a graphical data flow. Let’s model the data flow.

- c. Launch **Eclipse**
- d. Navigate to your *application component* folder by following **Data Sources** → **File** → **Flat-File** → **ZXXX** → **ZXXX###** (name: **Student ###**) and choose **New** → **DataSource** in the context menu (right-click).

Note: When working with the exercise, please replace in all names XXX with the ID of your course, and ### with your user number (three digits) provided by your instructor.

- e. **Next** → **Next**
- f. On the next screen use for the technical name **XXXDSMD###** (please replace **XXX** with the ID of your course and **###** with your number) and choose the DataSource Data Type Master Data Attributes.
- g. **Description:** **XXX### Material Master Data.**



New DataSource

Confirmation Page

BW Project: BW4_900_bwuser-001_en

Source System: PC_FILE

Source System Type: FILE

Application Component*: ZAA1001 Browse...

☐ Add to Favorites

Name*: AA1DSMD001

Description: AA1001 Material Master Data

DataSource Type*: Master Data Attributes

? < Back Next > **Finish** Cancel

Figure 5

- h. Finish
- i. Go to the **Extraction** tab, then choose the following settings:
 - i. Delta Process: No delta, only Full
 - ii. Direct Access: Allowed
 - iii. Adapter: Load Text-Type File from Local Workstation
 - iv. File Name: (browse for your *Product.CSV* file)
 - v. Ignore Header Rows: 1
 - vi. Data Format: Separated with Separator (for Example: CSV)
 - vii. Data Separator: ;

Extraction: AB1DSMD001 Active Version

Extraction Properties

Delta Process: No delta, only Full Delta Properties

Direct Access: Allowed

Streaming: Streaming cannot be triggered by this DataSource

Adapter: Load Text-Type File from Local Workstation

Extractor-Specific Properties

File Name: C:\Users\Nitin Kale\Google Drive\ITP 487\Fall 2019\Homework\ Browse...

Ignore Header Rows: 1

System Codepage: 4103 UTF-16LE Unicode / ISO/IEC 10646

☐ Specify Codepage to Replace System Codepage

Show File Content... Derive Fields from File...

Data Format

Data Format: Separated by Separator (Example: CSV)

Data Separator: ; ☐ Hex Browse...

Escape Sign: " ☐ Hex Browse...

Conversion Properties

☐ Set Conversion Language

☐ Set Number Format Thousands Separator: ☐ Decimal Point Separator: ☐

Figure 6

- j. Click on *Derive Fields from File* to propose the fields of the DataSource.
- k. Next → Finish.
- l. Click on tab **Fields** you will accept the proposed fields (no changes needed).

Fields: AB1DSMD001 Active Version

Fields

type filter text

Field	Key	Tr...	Data Type	Length	Field Type
[MATERIALDESCR] MaterialDescr		☒	CHAR	40	
[MATERIALNUMBER] MaterialNumber		☒	CHAR	8	
[PRODUCTCATEGORY] ProductCategory		☒	CHAR	3	
[DIVISION] Division		☒	CHAR	2	
[COLOR] Color		☒	CHAR	7	
[COMPONENTS] Components		☒	CHAR	16	

Add InfoObjects... Add Field Remove Up Down

General

Name: MATERIALDESCR

Description: MaterialDescr

InfoObject Template

InfoObject: Browse...

Data Type

Data Type: CHAR Length: 40

Technical Properties

Unit/Currency Field

Fixed Unit/Currency

☐ Key Field

☒ Case-Sensitive

Conversion Properties

Format: External

Conversion Routine: Browse...

Source Conv. Rout.:

External Length: 40

Selection Properties

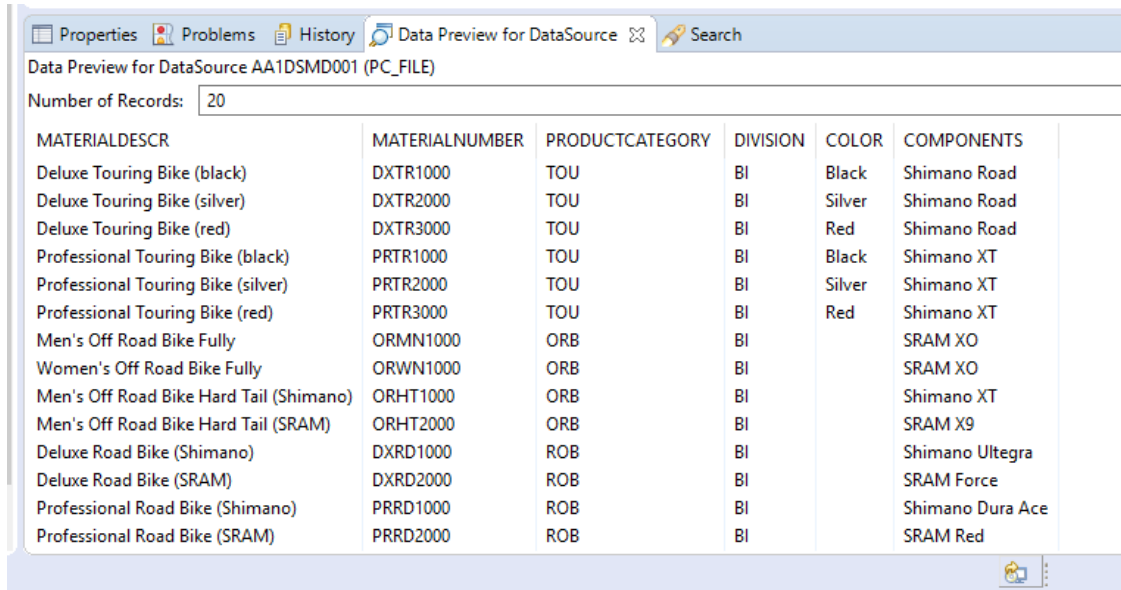
Selection: No Selection Possible

Selection Options: Undefined Selection Options

Figure 7

Question 1: Explain why we need 6 fields for material (shown above). *Hint: Refer to Exercise 3.2 – Modeling Basics.*

- m. Save and activate your DataSource.
- n. Test your DataSource on the icon **Show Data Preview**.
- o. The result should look like the following table:



MATERIALDESCR	MATERIALNUMBER	PRODUCTCATEGORY	DIVISION	COLOR	COMPONENTS
Deluxe Touring Bike (black)	DXTR1000	TOU	BI	Black	Shimano Road
Deluxe Touring Bike (silver)	DXTR2000	TOU	BI	Silver	Shimano Road
Deluxe Touring Bike (red)	DXTR3000	TOU	BI	Red	Shimano Road
Professional Touring Bike (black)	PRTR1000	TOU	BI	Black	Shimano XT
Professional Touring Bike (silver)	PRTR2000	TOU	BI	Silver	Shimano XT
Professional Touring Bike (red)	PRTR3000	TOU	BI	Red	Shimano XT
Men's Off Road Bike Fully	ORMN1000	ORB	BI		SRAM XO
Women's Off Road Bike Fully	ORWN1000	ORB	BI		SRAM XO
Men's Off Road Bike Hard Tail (Shimano)	ORHT1000	ORB	BI		Shimano XT
Men's Off Road Bike Hard Tail (SRAM)	ORHT2000	ORB	BI		SRAM X9
Deluxe Road Bike (Shimano)	DXRD1000	ROB	BI		Shimano Ultegra
Deluxe Road Bike (SRAM)	DXRD2000	ROB	BI		SRAM Force
Professional Road Bike (Shimano)	PRRD1000	ROB	BI		Shimano Dura Ace
Professional Road Bike (SRAM)	PRRD2000	ROB	BI		SRAM Red

Figure 8

2. Create Data Flow for Material

- a. Open **Eclipse**
- b. Navigate to your InfoArea folder **XX** → **XXX** → **XXX###** (name: **Student ###**).
- c. Choose **New** → **Data Flow Object** in the context menu (right click) of your InfoArea.

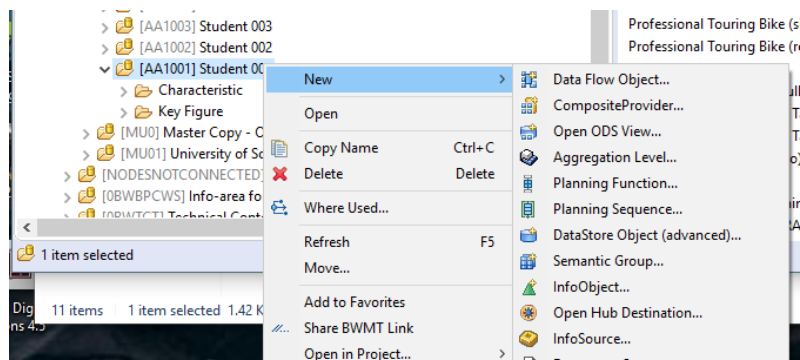


Figure 9

- d. For the technical name use **XXXDFMD###** and as description **XXX### Data Flow Material Master Data**.

New Data Flow Object

Data Flow Object

BW Project:* BW4_900_bwuser-001_en Browse...

InfoArea:* AA1001 Browse...

☐ Add to Favorites

Name:* AA1DFMD001

Description: AA1001 Data Flow Material Master Data

Copy From: Browse...

? Finish Cancel

Figure 10

- e. **Finish.**
- f. **Save and activate** your Data Flow.

3. Add the *Material* InfoObject to the Data Flow

- a. Open **Eclipse**
- b. Navigate to your InfoArea folder **XX** → **XXX** → **XXX###** (name: **Student ###**). Open your Data Flow from previous step with technical name **XXXDFMD###**.
- c. Drag and drop the **InfoObject** from the **BW Object Types** (on the right) into your Data Flow canvas.

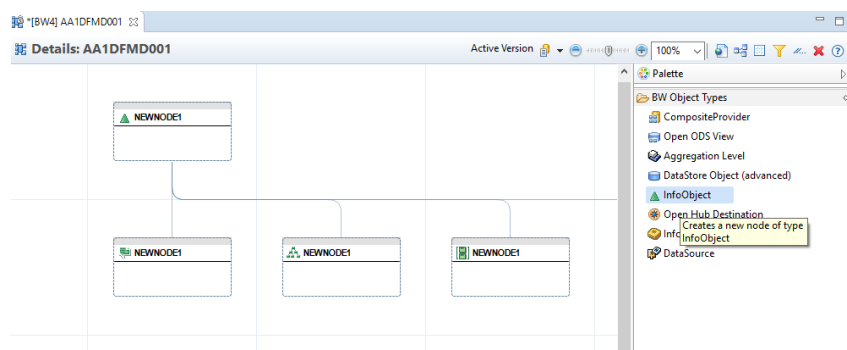


Figure 11

Note: The system inserts a top node representing the InfoObject itself and three or four sub-nodes for attributes, texts, hierarchies etc. We will not make use of hierarchies here and come back to this concept in a subsequent assignment.

- d. To insert your material InfoObject, choose **Add Object** in the context menu of the top node.

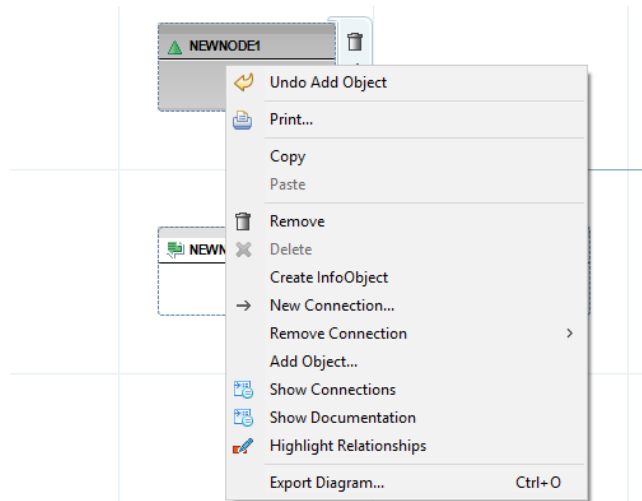


Figure 12

- e. Search and select your InfoObject with the technical name **XXXMA###**.

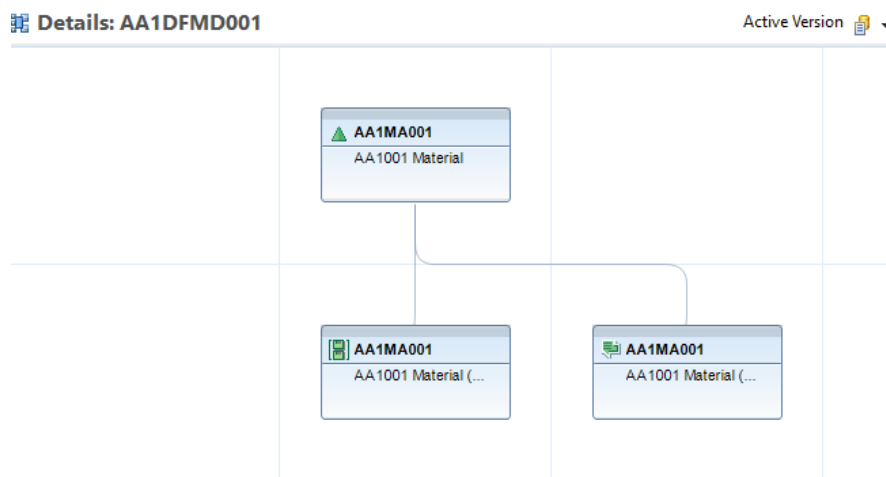


Figure 13

- f. The lower nodes should automatically get updated to Material Attributes and Material Texts.
- g. Then **save** and **activate** your Data Flow.

Question 2: Why did the system not add a Hierarchy node? Hint: Refer to Exercise 3.2.

4. Add the *Material* master data DataSource to Data Flow.

- Open **Eclipse**
- Navigate to your InfoArea folder **XX** → **XXX** → **XXX###** (name: **Student ###**). Open your Data Flow from previous step with technical name **XXXDFMD###**.
- Similar to the InfoObject in previous step, add a **DataSource** node to the Data Flow.

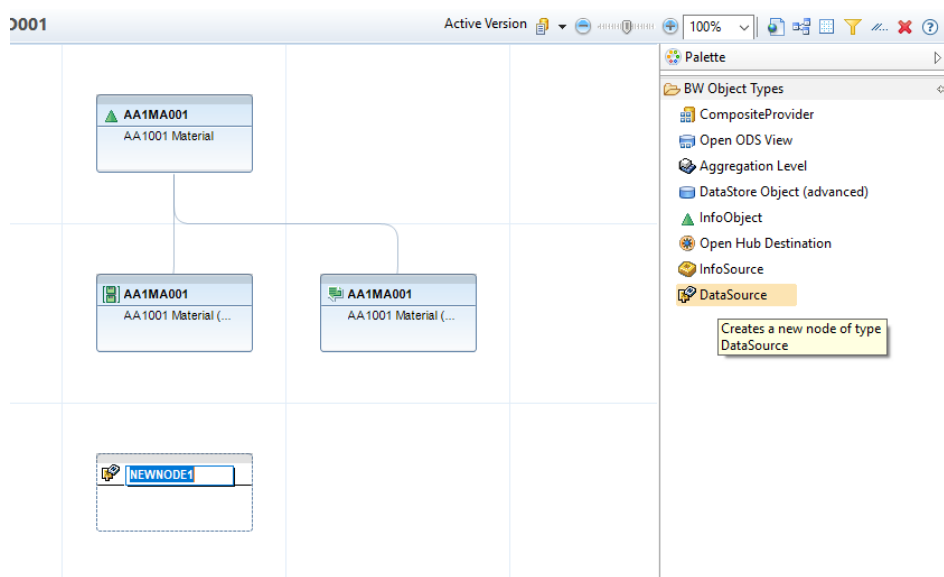


Figure 14

- To insert your DataSource, choose **Add Object** in the context menu of the datasource node.

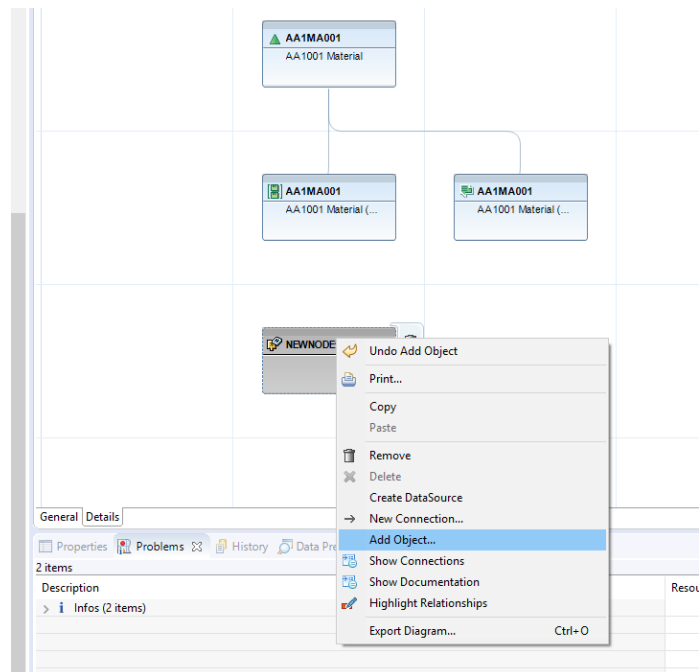


Figure 15

- e. Search and select your Data Source with the technical name **XXXDSMD###**.

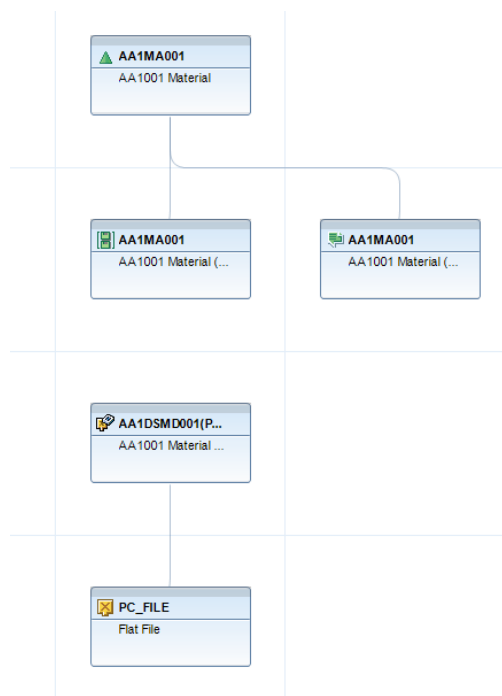


Figure 16

- f. **Save** and **activate** your Data Flow.

Your Data Flow now consists of a *source* and a *target*, you need to connect via a transformation in the next step.

5. Create and define rules for a transformation between the Data Source and the attributes of the InfoObject.

A *transformation* defines the rule set, how the fields are mapped between a source structure and a target structure.

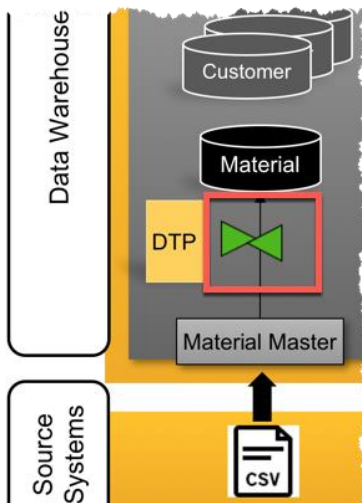


Figure 17

- Open **Eclipse**
- Navigate to your InfoArea folder **XX** → **XXX** → **XXX###** (name: **Student ###**). Open your Data Flow from previous step with technical name **XXXDFMD###**.
- Create a transformation from your data source to the attributes of your master data InfoObject. You do this by dragging the arrow **Drag to create a connection**. And connect to your Material Attributes.

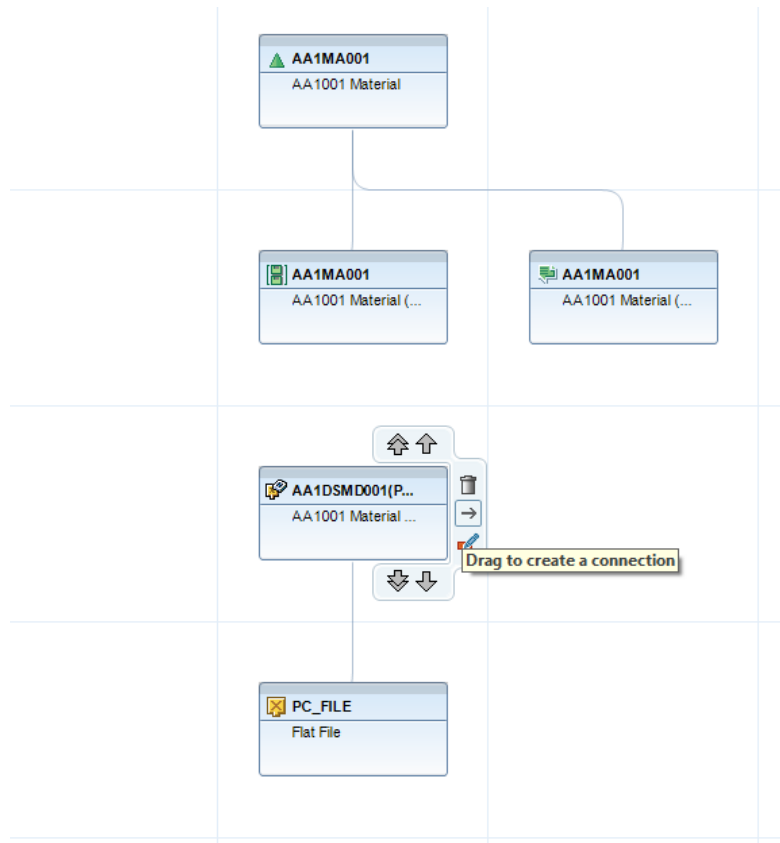


Figure 18

- d. A transformation is shown with a **bowtie** icon .
- e. You should now have a connection between data source node (XXX### Material Master Data) and attributes sub-node (XXX### Material (Attributes)).

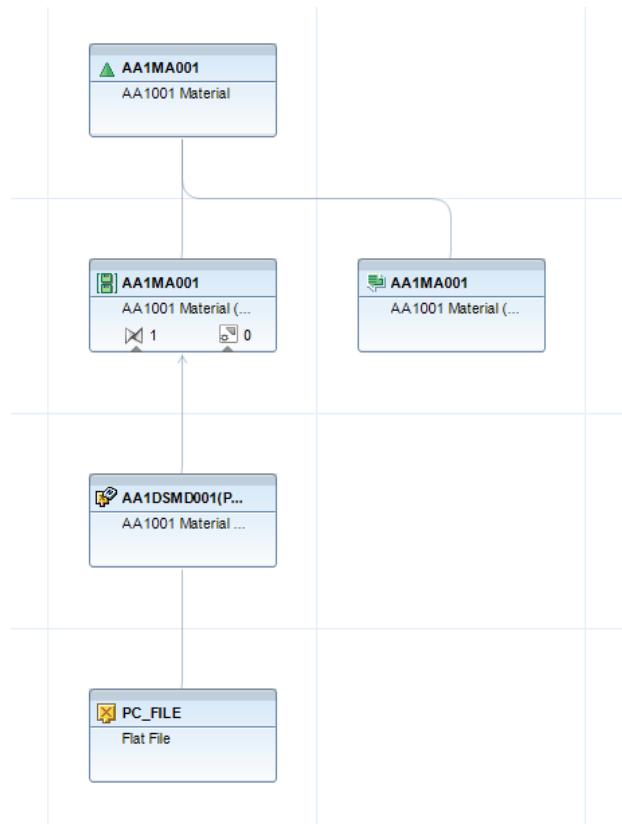
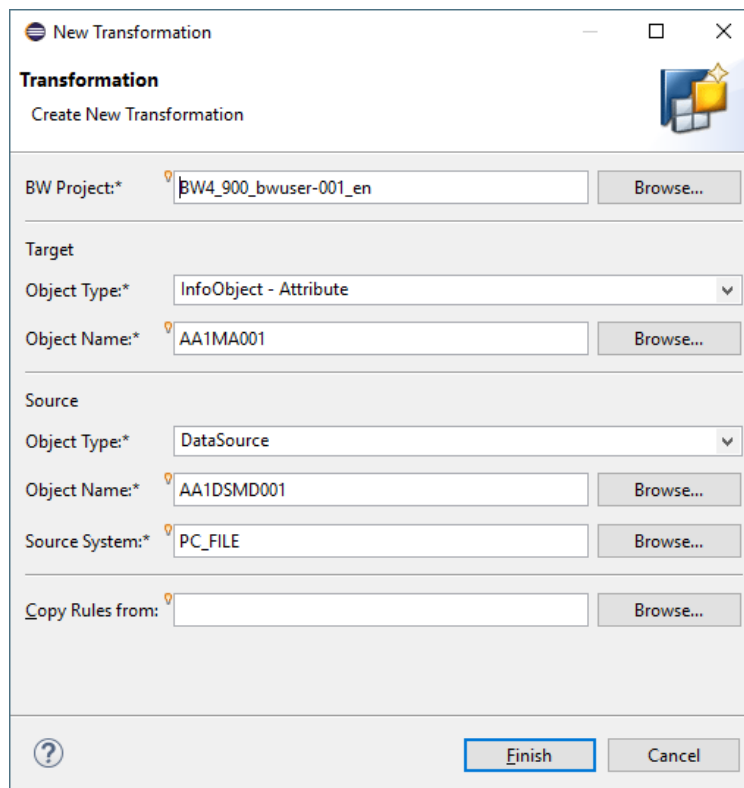


Figure 19

f. Then, **save** the Data Flow.

g. Double click on the white **transformation** icon .



New Transformation

Transformation
Create New Transformation

BW Project:* BW4_900_bwuser-001_en Browse...

Target

Object Type:* InfoObject - Attribute

Object Name:* AA1MA001 Browse...

Source

Object Type:* DataSource

Object Name:* AA1DSMD001 Browse...

Source System:* PC_FILE Browse...

Copy Rules from: Browse...

Finish Cancel

Figure 20

- h. **Finish.**
- i. Then click on the **Rules** tab.
- j. In the transformation editor, you map source fields (DataSource fields on the left side) to target fields (InfoObject fields on the right side). In our case we will just “move” the data without any transformation. All target fields need a corresponding source field!
- k. **Drag and drop** source fields to target fields as shown below in the mapping figure.

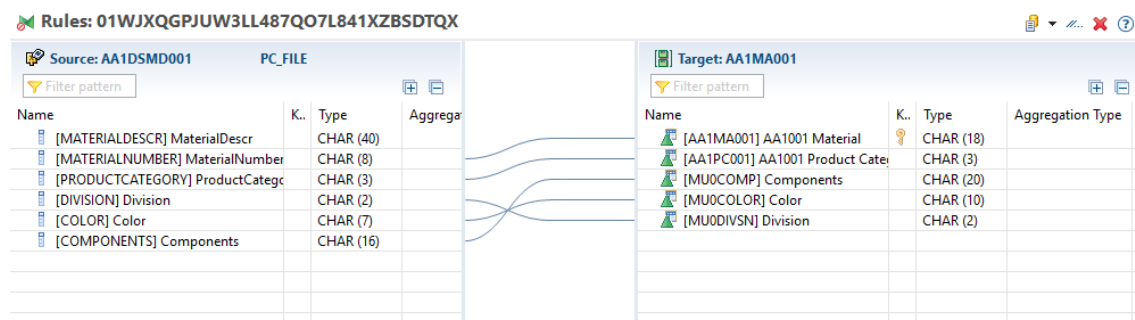


Figure 21

- l. **Save** and **activate** the transformation.
- m. Close the **Data Flow Object**.
- n. The transformation icon should now be green, so the transformation is active.
- o. **Save** and **activate** the Data Flow.

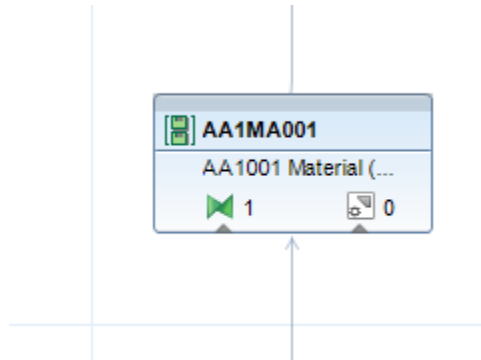


Figure 22

Question 3: In the mapping (rules) Figure 21 above, explain why the left table and right table have different icons for the fields.

Question 4: Explain why the Material Description is not connected to anything.

6. Create data transfer process and load data.

- a. Open **Eclipse**.
- b. The transfer of data (loading of ETL) between a source object and a target object is done by Data Transfer Processes (DTP), which can be understood as a program that is scheduled and actually “does” the data transfer.

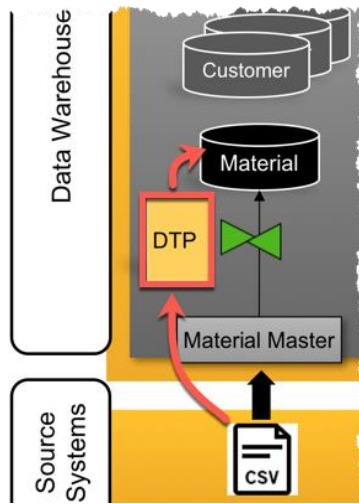


Figure 23

- c. Open the **Data Flows** tree and navigate to your InfoArea folder **XX** → **XXX** → **XXX###** (name: **Student ###**). Open your Data Flow from previous step with technical name **XXXDFMD###**.

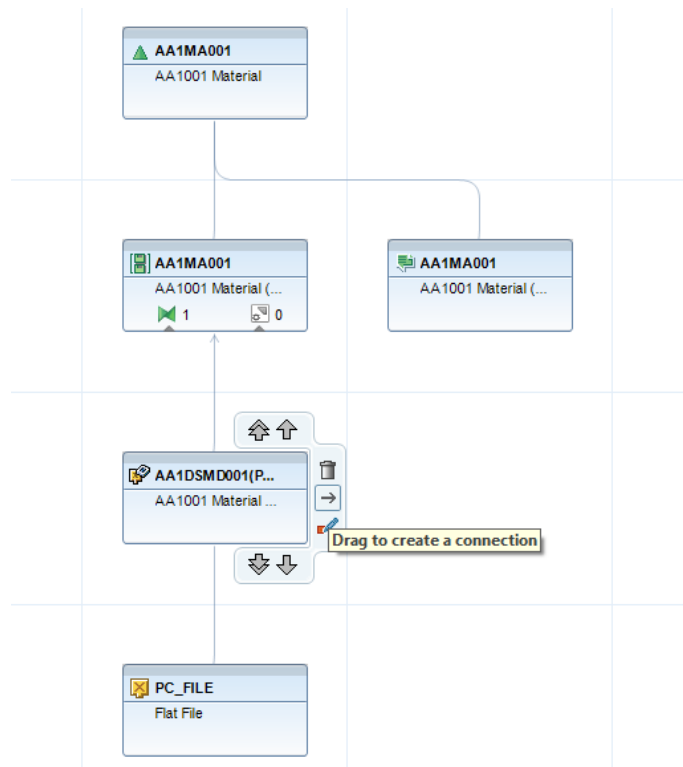


Figure 24

- d. **Drag a second connection** between data source (XXX### Material Master Data) and attributes sub-node (XXX### Material (Attributes)). Use the *arrow* to make the connection.
- e. Then, **save** the Data Flow

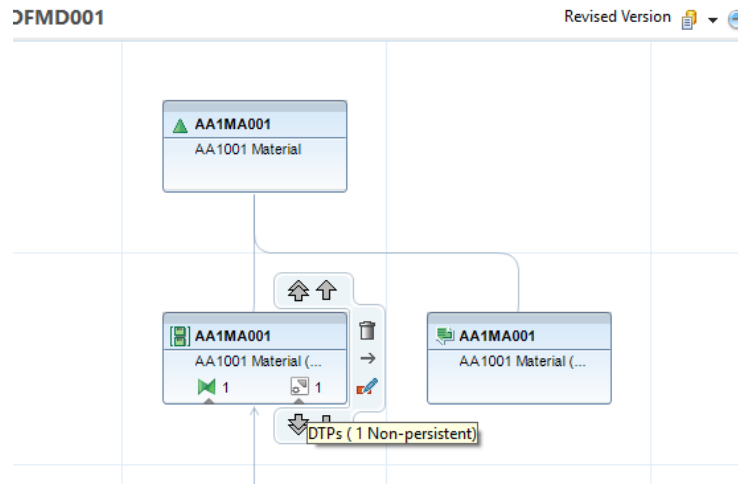


Figure 25

- f. Double click on the grey **DTP** icon.

New Data Transfer Process

Data Transfer Process
Create New Data Transfer Process

BW Project:* BW4_900_bwuser-001_en Browse...

Target

Object Type:* InfoObject - Attribute

Object Name:* AA1MA001 Browse... Browse Paths...

Source

Object Type:* DataSource

Object Name:* AA1DSMD001 Browse...

Source System:* PC_FILE Browse... Browse Paths...

Copy From: Browse...

? < Back Next > Finish Cancel

Figure 26

- g. **Next**
- h. **Finish**
- i. In the **DTP editor** make the following selections:
 - i. **General Tab**
 - (1) Extraction mode: **Full**
 - ii. **Extraction Tab**
 - (1) Adapter: **Load Text-Type File from Local Workstation**
 - (2) File Name: (the path to your *Product.csv* file)
 - (3) Ignore Header Rows: **1**

Adapter Selection

Adapter: Load Text-Type File from Local Workstation

Adapter Settings

File Name: uilinging\Data\Master Data\Product.csv Browse...

Ignore Header Rows: 1

System Codepage: 4103 UTF-16LE Unicode / ISO/IEC 10646

☐ Specify Codepage to Replace System Codepage

Data Format

Data Format: Separated by Separator (Example: CSV)

Data Separator: ; ☐ Hex

Escape Sign: " ☐ Hex

Figure 27

- j. **Save and Activate.**
- k. Click on **Start Execution** of Data Transfer Process.

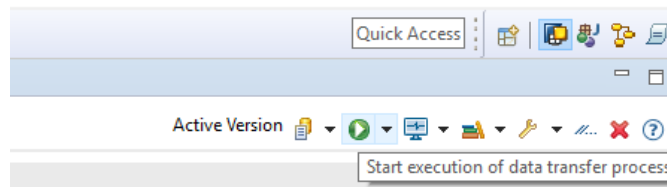


Figure 28

- l. You may get a system message about the server. Click on Continue.

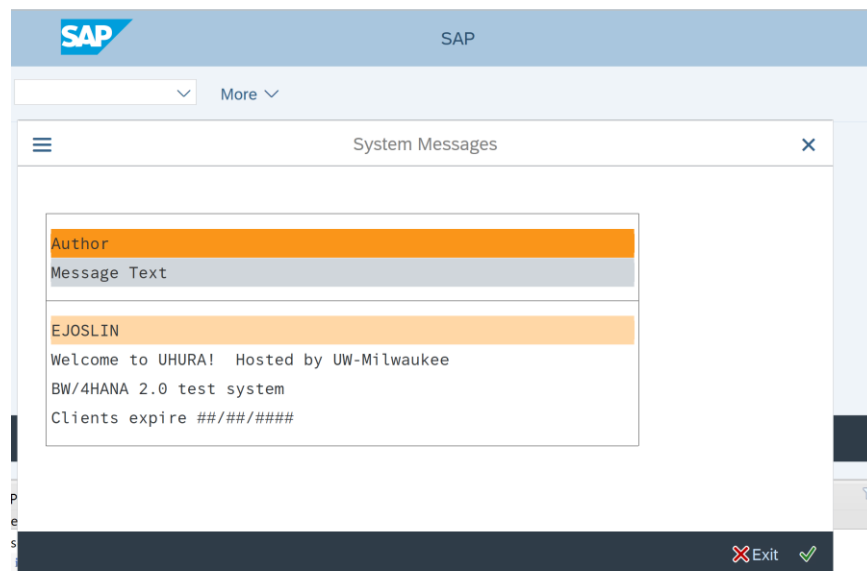


Figure 29

Processing Mode: Serial Partial SAP HANA Execution in Dialog Process		Execute
Program Flow		Breakpoints
✓ RSDS MU0DSMD FILE -> IOBJ ATTR MU0MA Fill data transfer intermediate storage Prepare for Extraction		Change Break
✓ Data Package Loop Extraction DataSource MU0 Material Master Data Filter Filter Out New Records with the Same Key RSDS MU0DSMD FILE -> IOBJ MU0MA Updating attributes for InfoObject MU0MA		Change Break

Figure 30

- m. Click **Execute**. Click **Continue** → **Allow** → **Yes**.
- n. If everything works fine, you see a success message, and can go to the monitor to see the log.

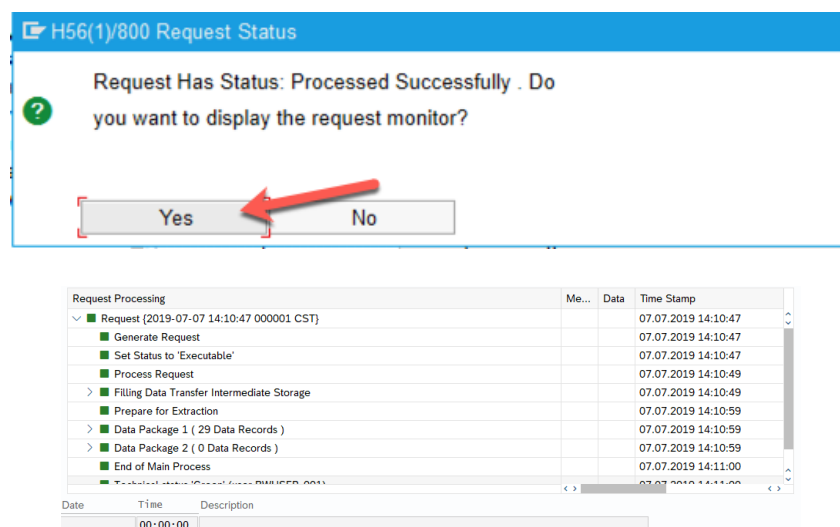


Figure 31

- o. You should see *Data Package 1 (29 Data Records)* message.
- p. Save and Activate your DTP tab.
- q. Close the **Execute Data Transfer Process** tab.
- r. **Save** and Activate your Data Flow.
- s. Your Material Attributes DTP should look like this (blue, not gray).

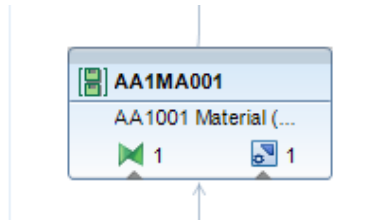


Figure 32

- t. You have now done the ETL for material *attributes* of your material characteristic. You need to ETL the material *texts* next.
- u. Close the DTP monitor tab and DTP tab.

Question 5: In this task, what part(s) of the E, T, L did you implement for material? Explain.

7. Verify the loaded attributes

- a. We want to verify that all the attributes of your material InfoObject have been migrated from the csv file to BW.
- b. Using the knowledge you have gained so far, you should explore and display
 - i. Product csv file showing the raw data
 - ii. Your own material master data that are now migrated into BW. Where would you look for this data?

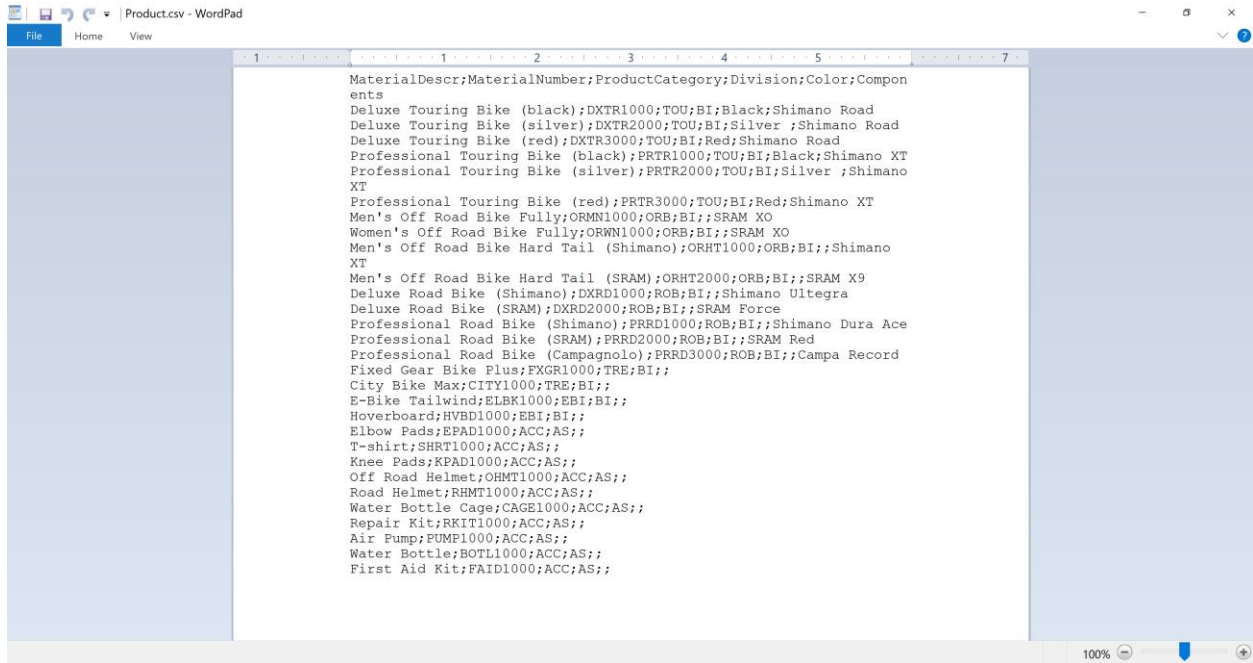


Figure 33

c. Verify, that the attributes are loaded (but no texts yet!).

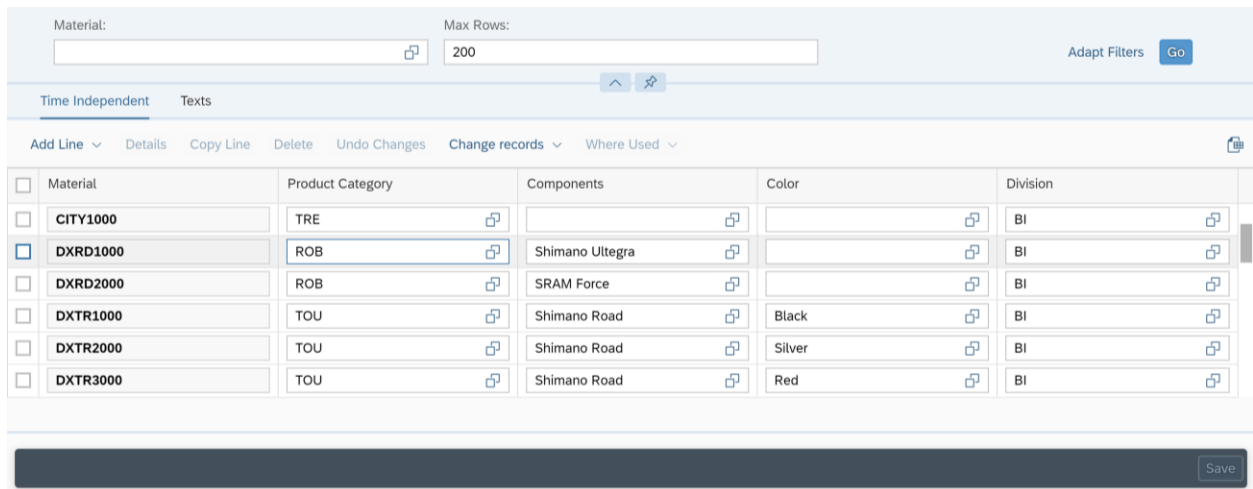


Figure 34

Question 6: Insert screenshot of product.csv file content.

Question 7: Insert screenshot of your material master data (attributes) from SAP BW/4HANA.

Question 8: Click on the Texts tab. Why are there no texts?

Question 9: Why Is there *medium* text but no short or any other text fields?

8. Create and define rules for a transformation for material text.

- Open **Eclipse**
- Navigate to your InfoArea folder **XX** → **XXX** → **XXX###** (name: **Student ###**). Open your Data Flow from previous step with technical name **XXXDFMD###**.

Note: We have texts and attributes in one data file product.csv. Can you identify which fields(s) are the texts? We do not need an additional data source!

- Create via **drag and drop** (arrow) a transformation from your data source (**XXX### Material Master Data**) to the descriptions (texts) (**XXX### Material (Texts)**) of your master data InfoObject.

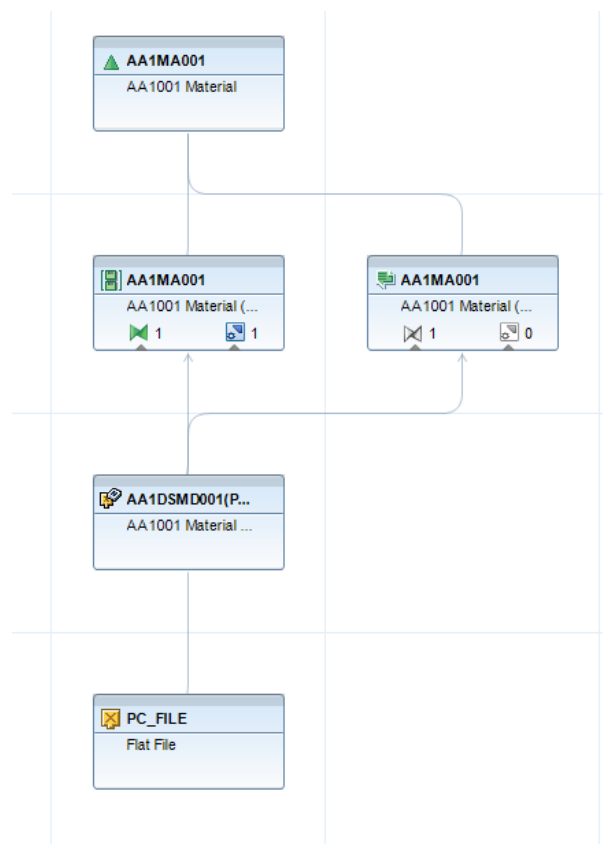


Figure 35

- Save** your Data Flow and double-click on the white **transformation icon**.

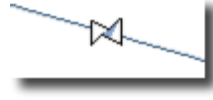


Figure 36

- e. **Finish.**
- f. Click on **Rules** tab.
- g. In the transformation editor, map your source fields (DataSource on the left side) to target fields (InfoObject on the right side):

DataSource	InfoObject
MATERIALDESCR	OTXTMD (Medium description)
MATERIALNUMBER	XXXMA### (XXX### Material)

- h. For **oLANGU** (Language key) double-click on the **Initial** icon.

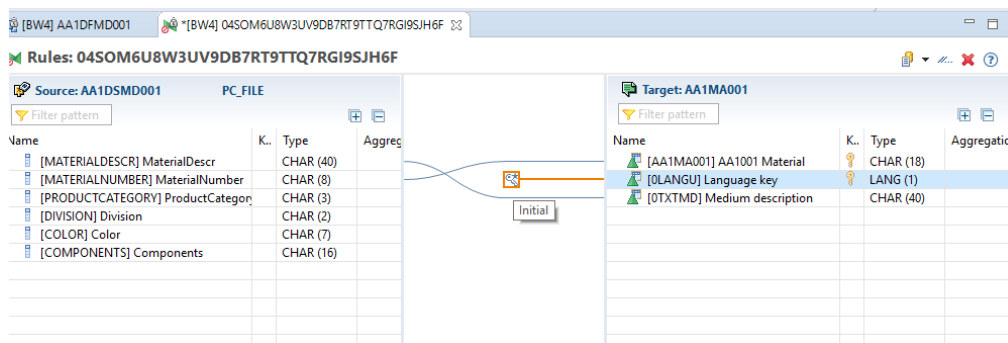
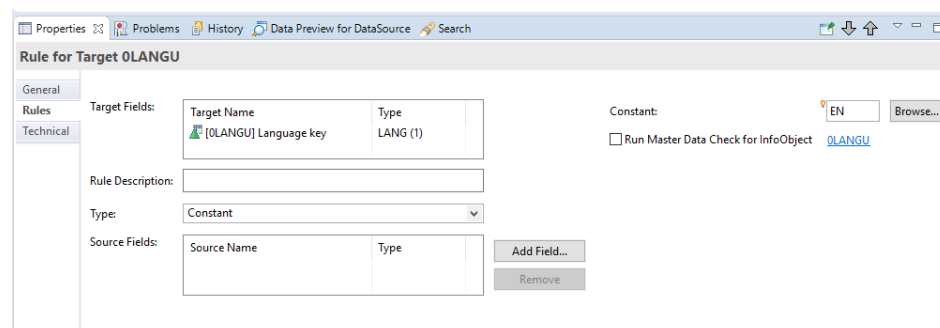


Figure 37

- i. Change the **Rule Type** to **Constant** as we know that the language of all texts is English, and type in for the Constant Value **EN**.

Note: This would not work if there were several languages!



- j. In a future project, you will introduce multilingual descriptions of material.

- k. **Save** and **activate** your transformation.
- l. Close **Rules**.

Question 10: In the mapping figure above, why are only material description and material number connected (and not the rest of the fields on the left side)?

Question 11: What rule are we implementing for the language key for material description?

9. Create a Data Transfer Process and load data.

- a. Open **Eclipse**
- b. Navigate to your InfoArea folder **XX** → **XXX** → **XXX###** (name: **Student ###**). Open your Data Flow from previous step with technical name **XXXDFMD###**.
- c. First, **drag a second connection** between data source (**XXX### Material Master Data**) and texts sub-node (**XXX### Material (Texts)**).

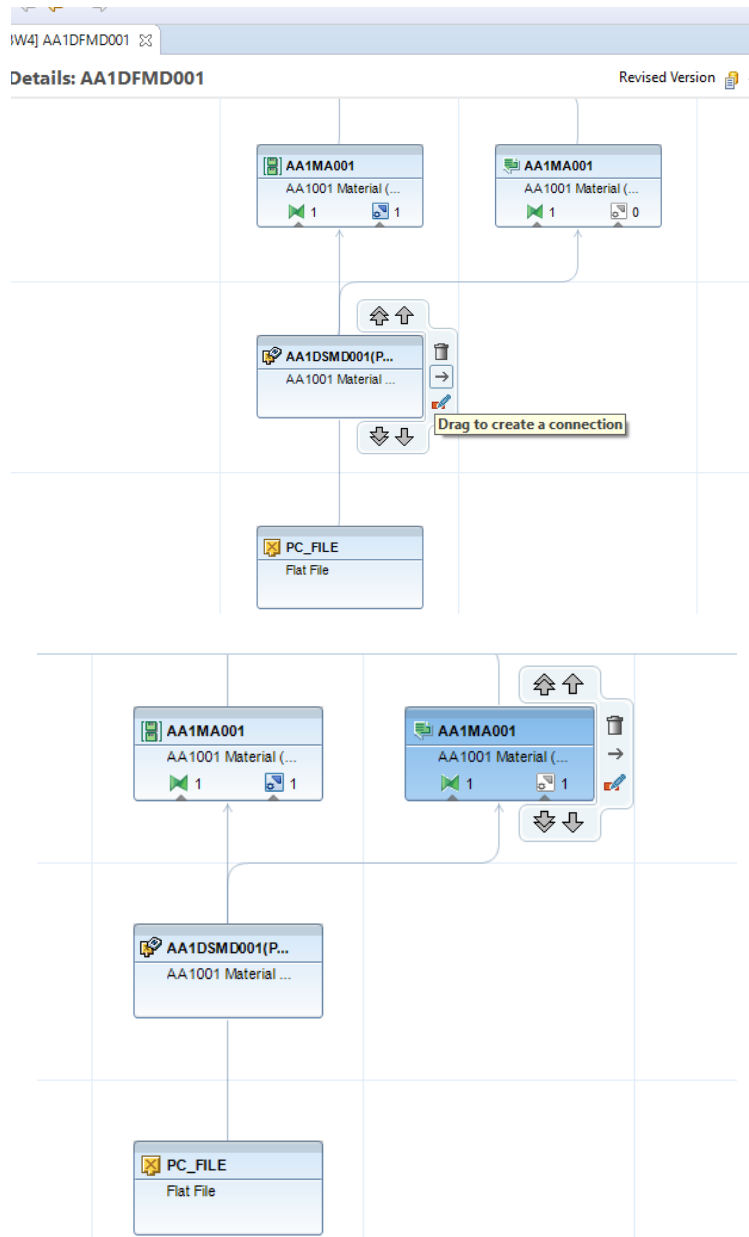


Figure 38

- d. Then, **save** the Data Flow and double-click on the grey **DTP** icon.
- e. **Next.**
- f. **Finish.**
- g. Change the inputs in the **DTP editor** as follows:
 - i. **General Tab**
 - (1) Extraction mode: **Full**
 - ii. **Extraction Tab**

- (1) Adapter: **Load Text-Type File from Local Workstation**
- (2) File Name: (the path to your *Product.csv* file)
- (3) Ignore Header Rows: **1**

Figure 39

- h. **Save and activate** the DTP
- i. Click **Start Execution of DTP**
- j. **Execute.**

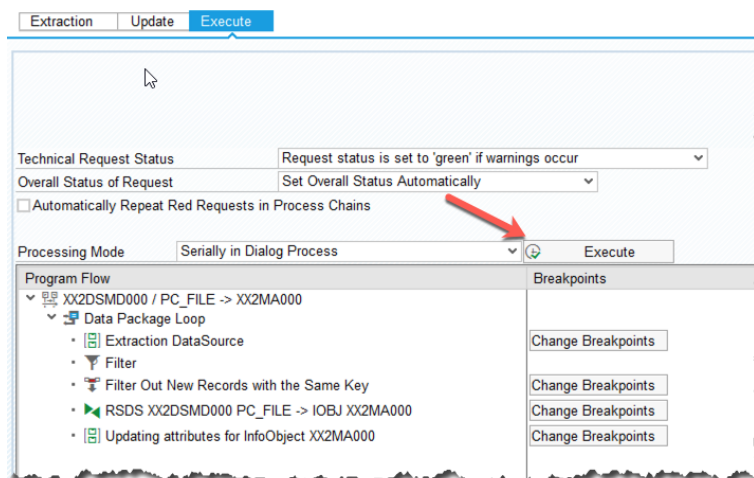


Figure 40

- k. If everything works fine, you see a success message, and can go to the monitor to see the log.
- l. **Allow** → **Yes**.
- m. Close the **Monitor** tab.
- n. Close the **Extraction** tab.
- o. **Save and activate** your Data Flow.
- p. Your Material Text DTP should be blue (not gray).

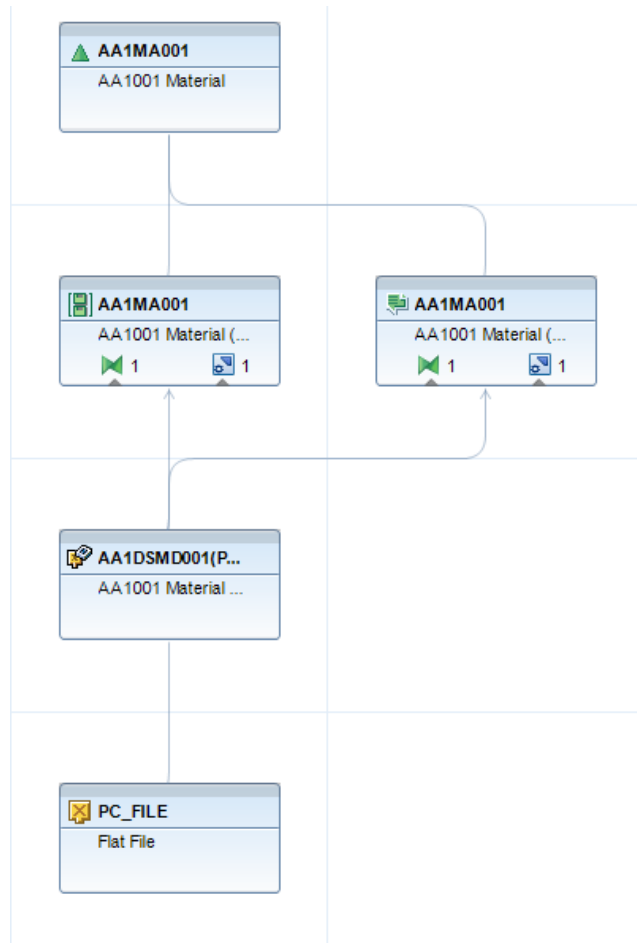
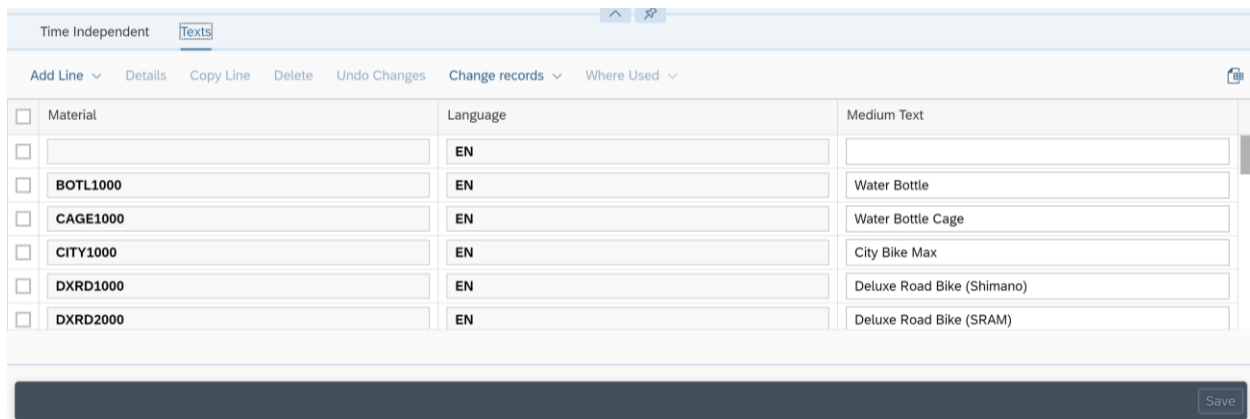


Figure 41

10. Verify the loaded texts

- On your own, verify that the material texts have been loaded correctly.



The screenshot shows the SAP Material Master - Texts table. The table has three columns: Material, Language, and Medium Text. The 'Material' column has a checkbox in the first row and material numbers in the others. The 'Language' column has 'EN' for all rows. The 'Medium Text' column has descriptions for each material. The table is part of a larger interface with a top bar 'Time Independent' and a menu bar with options like 'Add Line', 'Details', 'Copy Line', 'Delete', 'Undo Changes', 'Change records', and 'Where Used'. A 'Save' button is at the bottom right.

Material	Language	Medium Text
☐	EN	
☐ BOTL1000	EN	Water Bottle
☐ CAGE1000	EN	Water Bottle Cage
☐ CITY1000	EN	City Bike Max
☐ DXRD1000	EN	Deluxe Road Bike (Shimano)
☐ DXRD2000	EN	Deluxe Road Bike (SRAM)

Now you know how to migrate data from a source system into the data warehouse's material characteristic!

Question 12: Are there any non-English material descriptions? Why?

Question 13: Why is the description Medium in length?

Question 14: Insert a screenshot of the csv file. Highlight the texts part of It.

Question 15: Insert a screenshot of your material master data. Highlight the text part of It.